01 / ENGINEERING AND MATHEMATICAL THEORY

3D rotations and matrices

EX30 · Linear algebra, right-hand rule and physical applications

Study order. Start with the underlying linear algebra, then move to its geometric interpretation and finally apply it to forces and reference frames. Read this guide before programming.

1. What changes when moving from 2D to 3D?

A three-dimensional vector has three components:

$$\boldsymbol{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}.$$

A 3D rotation preserves the vector length and its origin, while changing its direction in space. Unlike a 2D rotation, a 3D rotation must specify *which axis* is used.

Rotating about x leaves the x component unchanged and transforms y, z. Rotating about y leaves y unchanged; rotating about z leaves z unchanged.

2. From planar rotations to the three fundamental matrices

Start with the 2D rotation studied in EX29, using $c = \cos\theta$ and $s = \sin\theta$:

$$\mathbf{R}_{2D}(\theta) = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}.$$

The construction rule is: **keep the rotation axis unchanged and insert the 2D rotation block into the other two components**.

Rotation about x: the y and z components change

$$\mathbf{R}_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{pmatrix}$$

$$v'_x = v_x, \qquad v'_y = c\,v_y - s\,v_z, \qquad v'_z = s\,v_y + c\,v_z.$$

Rotation about y: the x and z components change

$$\mathbf{R}_y(\theta) = \begin{pmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{pmatrix}$$

$$v'_x = c\,v_x + s\,v_z, \qquad v'_y = v_y, \qquad v'_z = -s\,v_x + c\,v_z.$$

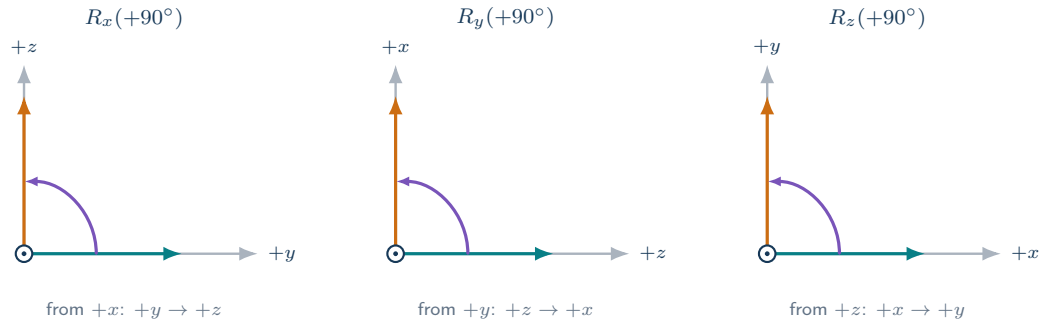
Rotation about z: the x and y components change

$$\mathbf{R}_z(\theta) = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$v'_x = c\,v_x - s\,v_y, \qquad v'_y = s\,v_x + c\,v_y, \qquad v'_z = v_z.$$

Signs matter. The sine entries in $\mathbf{R}_y$ do not appear in the same row positions as those in $\mathbf{R}_x$ and $\mathbf{R}_z$. Their signs follow the right-hand rule, not an arbitrary choice.

3. Visual reference: positive rotation direction



Each panel is viewed from the positive rotation axis towards the origin. The purple arc indicates positive rotation (right-hand rule).

The drawings illustrate an *active rotation* (the vector turns while the global axes remain fixed). Thus, a $+90°$ turn around x maps $+y$ to $+z$; around y it maps $+z$ to $+x$; and around z it maps $+x$ to $+y$.

4. Worked examples: matrix, vector, product

$R_x(+90°)$ **example.** For $\boldsymbol{b} = (1, 2, 0)^T$, $c = 0$ and $s = 1$:

$$\mathbf{R}_x(90°)\boldsymbol{b} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

$R_y(+90°)$ **example.** For $\boldsymbol{c} = (1, 0, 2)^T$:

$$\mathbf{R}_y(90°)\boldsymbol{c} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}.$$

$R_z(-90°)$ **example.** For $\boldsymbol{d} = (0, 3, 1)^T$, $c = 0$ and $s = -1$:

$$\mathbf{R}_z(-90°)\boldsymbol{d} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}.$$

5. Mathematical properties and why they hold

(i) Orthogonality. Every $\mathbf{R}$ has mutually perpendicular unit columns. For example, the columns of $\mathbf{R}_z$ are $(c, s, 0)^T$, $(-s, c, 0)^T$ and $(0, 0, 1)^T$. Their pairwise dot products vanish and their norms equal one because $c^2 + s^2 = 1$. Therefore:

$$\boxed{\mathbf{R}^T\mathbf{R} = \mathbf{I}_3, \qquad \mathbf{R}^{-1} = \mathbf{R}^T.}$$

(ii) Norm preservation. If $\boldsymbol{v}' = \mathbf{R}\boldsymbol{v}$:

$$\|\boldsymbol{v}'\|_2^2 = (\mathbf{R}\boldsymbol{v})^T(\mathbf{R}\boldsymbol{v}) = \boldsymbol{v}^T\mathbf{R}^T\mathbf{R}\boldsymbol{v} = \boldsymbol{v}^T\boldsymbol{v} = \|\boldsymbol{v}\|_2^2.$$

(iii) Determinant. These rotations have $\det \mathbf{R} = +1$ and preserve spatial orientation and volume. **Warning:** determinant $+1$ alone does not imply norm preservation: orthogonality must be checked.

6. Rotation order matters in 3D

Consider $\boldsymbol{p} = (1, 0, 0)^T$ and $+90°$ turns about x and z. First x, then z:

$$\mathbf{R}_z(90°)\mathbf{R}_x(90°)\boldsymbol{p} = \mathbf{R}_z(90°) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

But first z, then x gives:

$$\mathbf{R}_x(90^\circ)\mathbf{R}_z(90^\circ)\boldsymbol{p} = \mathbf{R}_x(90^\circ)\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\0\\1\end{pmatrix}.$$

These differ, so **3D rotation matrices generally do not commute**:

$$\boxed{\mathbf{R}_z\mathbf{R}_x \neq \mathbf{R}_x\mathbf{R}_z.}$$

In $\mathbf{R}_2\mathbf{R}_1\boldsymbol{v}$, $\mathbf{R}_1$ acts first and $\mathbf{R}_2$ second. This rule was introduced in EX29 and is essential in 3D.

7. Engineering application and inverse transformation

Suppose an initial force $\boldsymbol{F}_0 = (10, 0, 0)^T$ N is rotated by $+90^\circ$ around the fixed global z axis and then by $+90^\circ$ about the fixed global x axis:

$$\boldsymbol{F}_G = \mathbf{R}_x(90^\circ)\mathbf{R}_z(90^\circ)\boldsymbol{F}_0 = \mathbf{R}_x(90^\circ)\begin{pmatrix}0\\10\\0\end{pmatrix} = \begin{pmatrix}0\\0\\10\end{pmatrix}\text{N}.$$

With a lever arm $\boldsymbol{r} = (0.2, 0, 0)^T$ m, the resulting torque is:

$$\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{F}_G = \begin{pmatrix}0\\-2\\0\end{pmatrix}\text{N m}.$$

The cross product reinforces EX27.

For a *passive* transformation, if the local axes are rotated relative to global axes using $\mathbf{R}$, while the physical force does **not** move:

$$\boldsymbol{F}_G = \mathbf{R}\boldsymbol{F}_L, \qquad \boxed{\boldsymbol{F}_L = \mathbf{R}^T\boldsymbol{F}_G}.$$

The transpose does not physically rotate the force: it changes the coordinate basis used to describe it, as developed in the EX29 supplementary note.

8. Connecting the theory to Python

1. Select an axis and angle; convert degrees to radians with `np.deg2rad`.
2. Compute c, s; construct `R_x`, `R_y` or `R_z` as a 3×3 array.
3. Apply the rotations with `@` in the correct physical order.
4. Check shape `(3, 3)`, norm preservation, $\mathbf{R}^T\mathbf{R} \simeq \mathbf{I}_3$, $\det \mathbf{R} \simeq 1$, and recovery using the transpose.

Final rules. Keep the rotation axis fixed and insert a 2D block into its perpendicular plane. Positive angles follow the right-hand rule. In 3D, **order matters**. The transpose is the inverse of an **orthogonal rotation**, not of an arbitrary matrix.